LOGISTICS OPTIMIZATION FOR DELIVERY ROUTES – FLIPKART

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OVERVIEW

The project aims to answer key business questions, uncover inefficiencies, and recommend actionable improvements based on data analysis using the logistics dataset by Flipkart.

KEY OBJECTIVES –

- Identifying the root causes of delivery delays (e.g., congestion, hub processing issues).
- Optimizing delivery routes for faster, more cost-efficient fulfillment.
- Improving shipment efficiency and agent performance using data-driven insights.

TASK 1: DATA CLEANING & PREPARATION

DATA INTEGRITY CHECK

- Verified uniqueness of Order_ID (Primary Key)
- No duplicate records found
- Checked for NULL values in Traffic_Delay_Min
- No missing values detected

Insight:
Data is complete and maintains high integrity

DATE STANDARDIZATION

- Converted all date columns to YYYY-MM-DD format

Insight:
Ensures consistency for time-based analysis

DATA VALIDATION

- Checked for invalid delivery records
- No cases where delivery occurred before order date

Insight:
Dataset is logically consistent

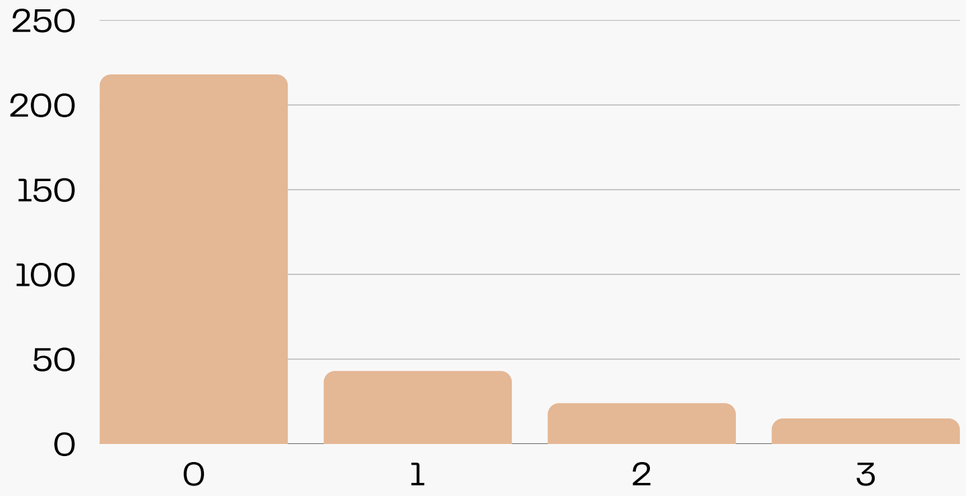
TASK 2: DELIVERY DELAY ANALYSIS

DELIVERY DELAY CALCULATION FOR EACH ORDER

```
SELECT Order_ID,
DATEDIFF(Actual_Delivery_Date,     Expected_Delivery_Date)     AS
delay_days
FROM flipkart_orders;
```

Order_ID	delivery_delay
FLP-ORD-0001	1
FLP-ORD-0002	0
FLP-ORD-0003	0
FLP-ORD-0004	0
FLP-ORD-0005	0
FLP-ORD-0006	1
FLP-ORD-0007	0
FLP-ORD-0008	0
FLP-ORD-0009	0
FLP-ORD-0010	3
FLP-ORD-0011	0
FLP-ORD-0012	1

Delivery Delay Distribution (in Days)



INSIGHT:

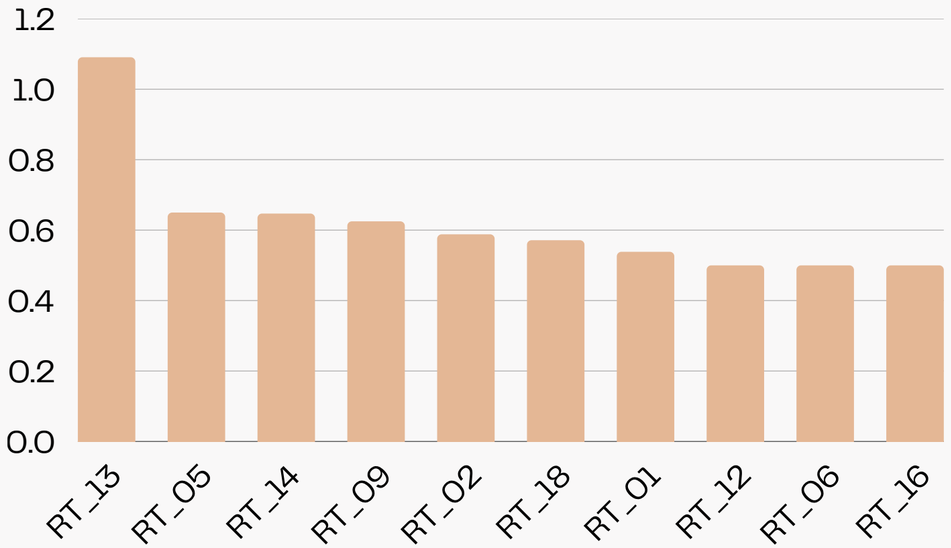
- Majority of deliveries are on-time (0 delay)
- A smaller portion shows 1–3 day delays
- Indicates inefficiencies in certain routes

TASK 2: DELIVERY DELAY ANALYSIS

TOP ROUTES WITH HIGHEST DELIVERY DELAY

```
SELECT Route_ID,
      AVG(DATEDIFF(Actual_Delivery_Date, Expected_Delivery_Date)) AS avg_delay
FROM flipkart_orders
GROUP BY Route_ID
ORDER BY avg_delay DESC LIMIT 10;
```

Route_ID	avg_delay
RT_13	1.0909
RT_05	0.6500
RT_14	0.6471
RT_09	0.6250
RT_02	0.5882
RT_18	0.5714
RT_01	0.5385
RT_12	0.5000
RT_06	0.5000
RT_16	0.5000



INSIGHT:

- Certain routes consistently show higher delays
- Indicates route inefficiencies or traffic congestion
- These routes require priority optimization

TASK 2: DELIVERY DELAY ANALYSIS

WAREHOUSE-WISE DELAY ANALYSIS

```
SELECT Warehouse_ID,
       DATEDIFF(Actual_Delivery_Date, Expected_Delivery_Date) AS
delay_days,
    RANK() OVER (
      PARTITION BY Warehouse_ID
                    ORDER BY DATEDIFF(Actual_Delivery_Date,
Expected_Delivery_Date) DESC
    ) AS delay_rank FROM flipkart_orders;
```

Order_ID	Warehouse_ID	delivery_delay	Warehouse_
FLP-ORD-0077	WH_01	3	1
FLP-ORD-0194	WH_01	3	1
FLP-ORD-0210	WH_01	2	3
FLP-ORD-0055	WH_01	2	3
FLP-ORD-0148	WH_01	1	5
FLP-ORD-0296	WH_01	1	5
FLP-ORD-0002	WH_01	0	7
FLP-ORD-0164	WH_01	0	7
FLP-ORD-0054	WH_01	0	7
FLP-ORD-0094	WH_01	0	7
FLP-ORD-0029	WH_01	0	7
FLP-ORD-0294	WH_01	0	7
FLP-ORD-0286	WH_01	0	7
FLP-ORD-0184	WH_01	0	7
FLP-ORD-0295	WH_01	0	7
FLP-ORD-0223	WH_01	0	7

INSIGHT:

- Some warehouses handle more delayed orders
- Indicates operational inefficiencies
- Helps identify bottleneck warehouses

TASK 3: ROUTE OPTIMIZATION INSIGHTS

ROUTE PERFORMANCE METRICS

```
SELECT r.Route_ID,
        AVG(DATEDIFF(o.Actual_Delivery_Date, o.Order_Date)) AS
avg_delivery_time,
    AVG(r.Traffic_Delay_Min) AS avg_traffic_delay,
    AVG(r.Distance_KM / r.Average_Travel_Time_Min) AS efficiency_ratio
FROM flipkart_orders o
JOIN flipkart_routes r
ON o.Route_ID = r.Route_ID
GROUP BY r.Route_ID;
```

Route_ID	avg_delivery_time	avg_traffic_delay	efficiency_ratio
RT_20	4.8000	37.0000	2.21590000
RT_14	4.8824	36.0000	1.00110000
RT_12	4.6111	58.0000	4.98450000
RT_10	4.4211	15.0000	4.94410000
RT_05	4.9000	55.0000	2.45500000
RT_19	4.0667	50.0000	7.48160000
RT_17	3.8000	81.0000	1.26640000
RT_11	4.0000	20.0000	2.24480000
RT_15	5.0588	87.0000	4.86810000
RT_13	4.7273	56.0000	0.72790000
RT_08	4.5000	90.0000	1.64100000
RT_03	4.3000	29.0000	1.12950000
RT_18	4.5714	17.0000	1.83890000
RT_06	4.4167	15.0000	1.92230000
RT_01	4.2308	67.0000	1.59900000
RT_04	4.0625	23.0000	1.97120000

INSIGHT:

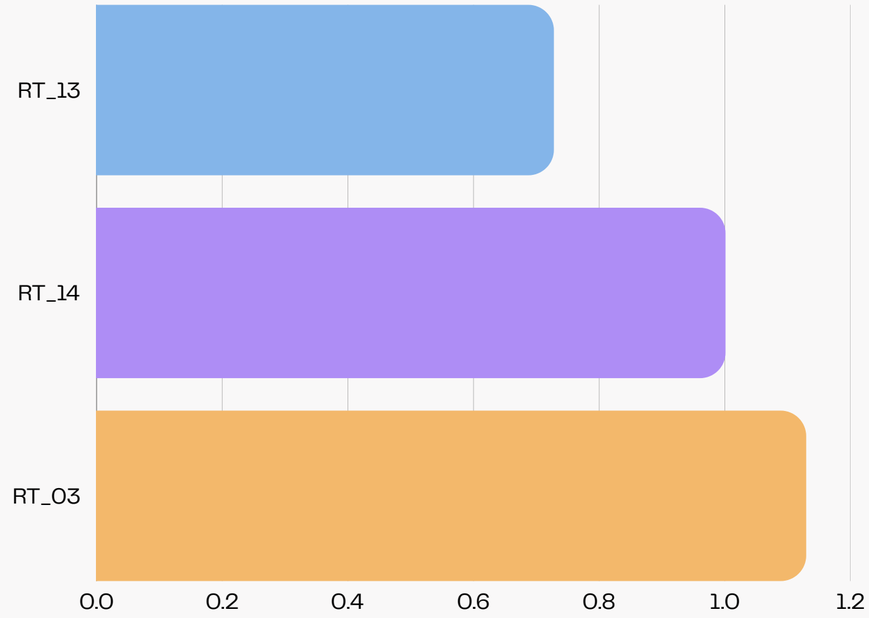
- Delivery time varies across routes
- Traffic delay directly impacts delivery efficiency
- Efficiency ratio helps identify route performance

TASK 3: ROUTE OPTIMIZATION INSIGHTS

WORST PERFORMING ROUTES(WORST 3)

```
SELECT Route_ID,
      Distance_KM / Average_Travel_Time_Min AS efficiency_ratio
FROM flipkart_routes
ORDER BY efficiency_ratio ASC
LIMIT 3;
```

Route_ID	efficiency_ratio
RT_13	0.72790000
RT_14	1.00110000
RT_03	1.12950000



Insight:

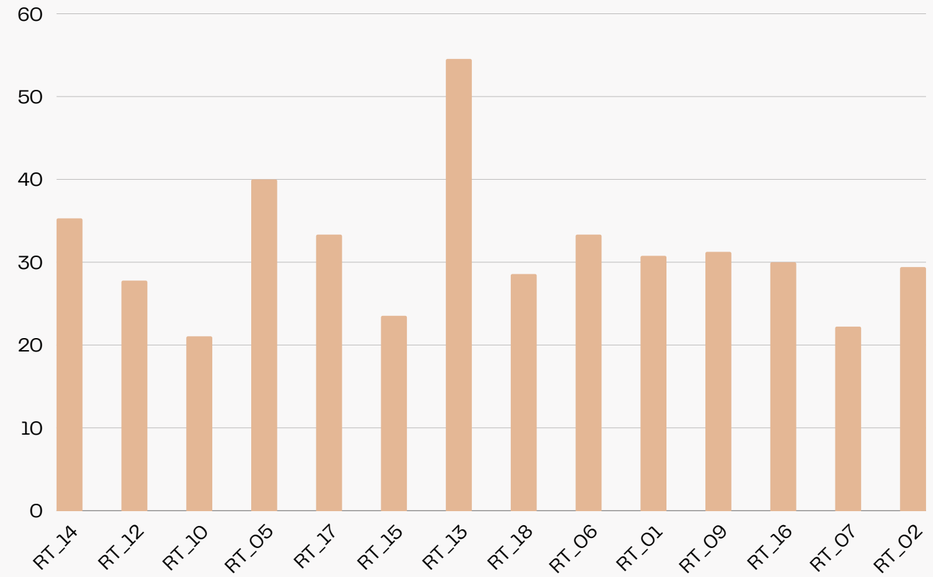
- Routes with low efficiency ratio indicate slow movement
- These routes are key candidates for optimization

TASK 3: ROUTE OPTIMIZATION INSIGHTS

ROUTES WITH HIGH DELAY %

```
SELECT Route_ID,
      (SUM(CASE
          WHEN Actual_Delivery_Date > Expected_Delivery_Date THEN 1
          ELSE 0
       END) * 100.0) / COUNT(*) AS delay_percentage
FROM flipkart_orders
GROUP BY Route_ID HAVING delay_percentage > 20;
```

Route_ID	Delayed_Shipments
RT_14	35.2941
RT_12	27.7778
RT_10	21.0526
RT_05	40.0000
RT_17	33.3333
RT_15	23.5294
RT_13	54.5455
RT_18	28.5714
RT_06	33.3333
RT_01	30.7692
RT_09	31.2500
RT_16	30.0000
RT_07	22.2222
RT_02	29.4118



INSIGHT:

- Many routes have more than 20% delayed deliveries
- Indicates serious inefficiencies
- Requires immediate optimization

RECOMMENDATIONS :

01 OPTIMIZE LOW-EFFICIENCY ROUTES

- Routes with low distance-to-time ratios indicate slower movement
- These routes should be redesigned using faster or alternate paths

02 MANAGE HIGH-TRAFFIC AREAS

- High traffic delay significantly impacts delivery timelines
- Implement traffic-aware scheduling and avoid peak hours

03 IMPROVE ROUTE PLANNING STRATEGIES

- High delay percentages indicate inefficient route planning
- Optimize delivery allocation and route sequencing

04 IDENTIFY CRITICAL BOTTLENECK ROUTES

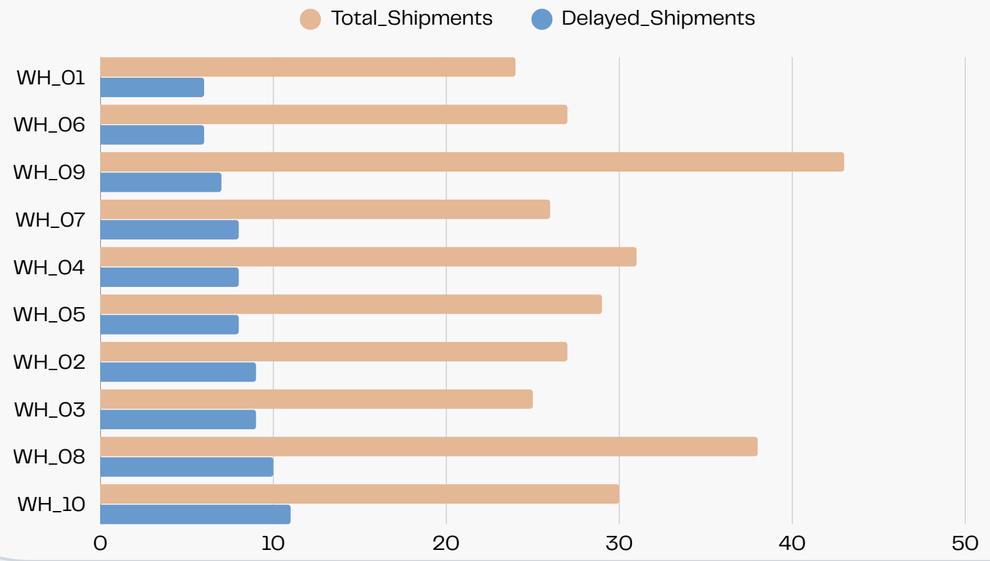
- Routes with low efficiency, high traffic delay, and high delay percentage represent critical bottlenecks.
- These routes should be prioritized for optimization.

TASK 4: WAREHOUSE PERFORMANCE

WAREHOUSE DELIVERY PERFORMANCE

```
SELECT Warehouse_ID,
      COUNT(*) AS total_shipments,
      SUM(CASE
          WHEN Actual_Delivery_Date > Expected_Delivery_Date THEN 1
          ELSE 0
        END) AS delayed_shipments
FROM flipkart_orders
GROUP BY Warehouse_ID;
```

Warehouse_ID	Total_Shipments	Delayed_Shipments
WH_07	26	8
WH_01	24	6
WH_04	31	8
WH_02	27	9
WH_09	43	7
WH_10	30	11
WH_08	38	10
WH_05	29	8
WH_06	27	6
WH_03	25	9



INSIGHT

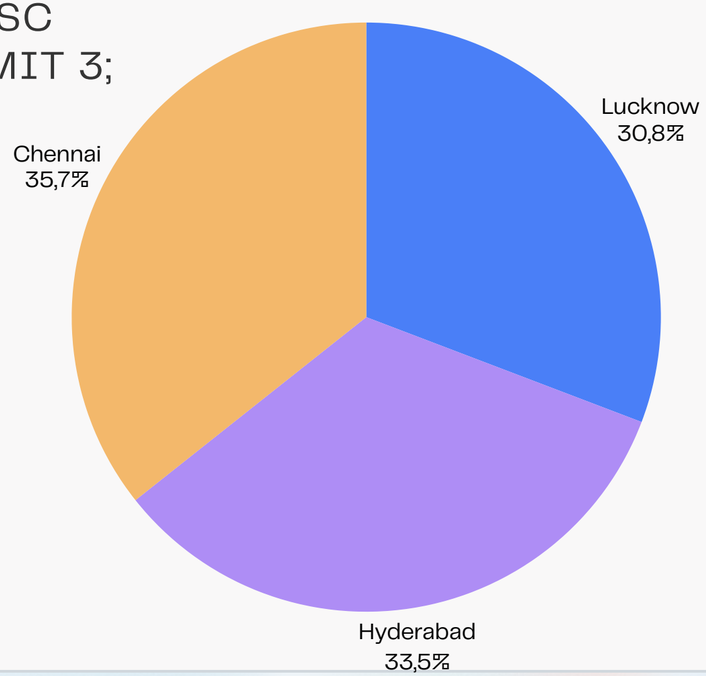
- Some warehouses show higher delayed shipments ratio
- Indicates inefficiency in processing or dispatch

TASK 4: WAREHOUSE PERFORMANCE

WAREHOUSE PERFORMANCE: PROCESSING TIME (TOP 3)

```
SELECT Warehouse_ID,
       Warehouse_Name,
       Average_Processing_Time_Min
FROM flipkart_warehouses
ORDER     BY     Average_Processing_Time_Min
DESC
LIMIT 3;
```

Warehouse_ID	Warehouse_Name	Average_Processing_Time_Min
WH_10	Flipkart Fulfillment Center Chennai	117
WH_09	Flipkart Fulfillment Center Hyderabad	110
WH_01	Flipkart Fulfillment Center Lucknow	101



INSIGHT

- Top warehouses have higher processing times
- Indicates potential operational delays
- These warehouses may act as bottlenecks

TASK 4: WAREHOUSE PERFORMANCE

IDENTIFYING BOTTLENECK WAREHOUSES

```
WITH avg_processing AS (
   SELECT AVG(Average_Processing_Time_Min) AS global_avg
   FROM flipkart_warehouses
)
SELECT Warehouse_ID,
      Warehouse_Name,
      Average_Processing_Time_Min
FROM flipkart_warehouses, avg_processing
WHERE Average_Processing_Time_Min > global_avg;
```

Warehouse_ID	Average_Processing_Time_Min
WH_01	101
WH_03	84
WH_04	81
WH_06	95
WH_09	110
WH_10	117

INSIGHT

- WAREHOUSES EXCEEDING GLOBAL AVERAGE PROCESSING TIME
- IDENTIFIED AS BOTTLENECKS
- REQUIRE OPERATIONAL IMPROVEMENT

KEY FINDINGS

01

Some warehouses have significantly higher processing times

02

Delayed shipments are concentrated in specific warehouses

03

Bottleneck warehouses impact overall delivery performance

TASK 5: DELIVERY AGENT PERFORMANCE

DELIVERY AGENT PERFORMANCE (ON-TIME %)

```
SELECT a.Agent_ID,
      a.Agent_Name,
      (SUM(CASE
            WHEN o.Actual_Delivery_Date <= o.Expected_Delivery_Date THEN 1
            ELSE 0
          END) * 100.0) / COUNT(*) AS on_time_percentage
FROM flipkart_orders o
JOIN flipkart_deliveryagents a
ON o.Agent_ID = a.Agent_ID
GROUP BY a.Agent_ID, a.Agent_Name;
```

Agent_ID	Route_ID	On_Time_Delivery_Percentage	Agent_Rank
AG_049	RT_01	97.2	1
AG_047	RT_01	81.2	2
AG_002	RT_01	73.2	3
AG_038	RT_02	91.6	1
AG_007	RT_02	85.9	2
AG_020	RT_02	85.1	3
AG_028	RT_02	81.7	4
AG_026	RT_02	72.2	5
AG_013	RT_03	94.1	1
AG_001	RT_03	86.7	2
AG_017	RT_04	90.4	1
AG_050	RT_04	79.7	2
AG_040	RT_05	93.4	1
AG_041	RT_05	80.2	2
AG_021	RT_06	83.5	1
AG_004	RT_06	73	2

Insight

- Significant variation in agent performance
- High-performing agents consistently meet delivery timelines
- Low-performing agents require attention

TASK 5: DELIVERY AGENT PERFORMANCE

TOP VS BOTTOM PERFORMING AGENTS

```
WITH agent_performance AS (
   SELECT o.Agent_ID, ROUND(SUM(CASE
            WHEN o.Actual_Delivery_Date <= o.Expected_Delivery_Date THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS on_time_pct
   FROM flipkart_orders o
   GROUP BY o.Agent_ID),
ranked AS (
   SELECT p.Agent_ID, a.Avg_Speed_KMPH, p.on_time_pct,
      RANK() OVER (ORDER BY p.on_time_pct DESC) AS top_rank,
      RANK() OVER (ORDER BY p.on_time_pct ASC) AS bottom_rank
   FROM agent_performance p
   JOIN flipkart_deliveryagents a
   ON p.Agent_ID = a.Agent_ID)
SELECT 'Top 5' AS Category, ROUND(AVG(Avg_Speed_KMPH), 2) AS avg_speed
FROM ranked WHERE top_rank <= 5
UNION ALL
SELECT 'Bottom 5', ROUND(AVG(Avg_Speed_KMPH), 2)
FROM ranked
WHERE bottom_rank <= 5;
```

Category	avg_speed
Top 5	42.79
Bottom 5	42.2

Insight:

Although top-performing agents exhibit slightly higher average speeds, the difference is minimal, suggesting that delivery performance is influenced more by external factors such as traffic conditions, route efficiency, and workload distribution rather than speed alone.

TASK 5: DELIVERY AGENT PERFORMANCE

AGENTS REQUIRING IMPROVEMENT

```
SELECT a.Agent_ID,
       a.Agent_Name,
       (SUM(CASE
            WHEN o.Actual_Delivery_Date <= o.Expected_Delivery_Date THEN 1
            ELSE 0
           END) * 100.0) / COUNT(*) AS on_time_percentage
FROM flipkart_orders o
INNER JOIN flipkart_deliveryagents a
ON o.Agent_ID = a.Agent_ID
GROUP BY a.Agent_ID, a.Agent_Name
HAVING (SUM(CASE
             WHEN o.Actual_Delivery_Date <= o.Expected_Delivery_Date THEN 1
             ELSE 0
            END) * 100.0) / COUNT(*) < 80;
```

Agent_ID	Agent_name	on_time_percentage
AG_033	Pooja Reddy	50.00000
AG_001	Arun Nair	66.66667
AG_028	Priya Nair	75.00000
AG_013	Rajesh Patel	50.00000
AG_020	Kiran Kumar	33.33333
AG_002	Vikram Nair	66.66667
AG_027	Priya Kumar	60.00000
AG_016	Anita Kumar	71.42857
AG_015	Priya Nair	66.66667
AG_036	Vikram Reddy	62.50000
AG_025	Kiran Kumar	75.00000
AG_048	Anita Reddy	66.66667
AG_018	Anita Patel	75.00000
AG_030	Arun Patel	60.00000
AG_045	Arun Gupta	57.14286
AG_035	Anita Patel	50.00000

Insight

- Agents below 80% on-time performance are underperforming
- Require training and monitoring

RECOMMENDATIONS :

01 TARGETED TRAINING & TIME MANAGEMENT

- Provide focused training on time management, scheduling, and prioritization
- Standardize best practices from top-performing agents

02 WORKLOAD BALANCING

- Distribute deliveries based on agent capacity and past performance
- Avoid overloading specific agents to reduce delays

03 ROUTE OPTIMIZATION SUPPORT

- Assign efficient routes and enable traffic-aware navigation
- Use optimized route planning to improve delivery efficiency

04 CONTINUOUS PERFORMANCE MONITORING

- Track KPIs like on-time delivery % and delay frequency
- Identify low performers early and take corrective actions

TASK 6: SHIPMENT TRACKING ANALYTICS

SHIPMENT TRACKING: LATEST CHECKPOINT

```
SELECT
    t.Order_ID, t.Checkpoint, t.Checkpoint_Time AS last_checkpoint
FROM flipkart_shipmenttracking t
JOIN (
    SELECT Order_ID, MAX(Checkpoint_Time) AS last_checkpoint
    FROM flipkart_shipmenttracking
    GROUP BY Order_ID) m
ON t.Order_ID = m.Order_ID
AND t.Checkpoint_Time = m.last_checkpoint;
```

Order_ID	Checkpoint	last_checkpoint
FLP-ORD-0095	Hub_1_Lucknow	24-08-2025 11:49
FLP-ORD-0074	Hub_3_Delhi	30-08-2025 00:35
FLP-ORD-0199	Hub_2_Pune	14-07-2025 10:41
FLP-ORD-0190	Hub_2_Pune	20-08-2025 22:15
FLP-ORD-0021	Hub_1_Mumbai	21-08-2025 18:46
FLP-ORD-0246	Hub_4_Jaipur	23-07-2025 09:31
FLP-ORD-0061	Hub_4_Bengaluru	25-08-2025 20:37
FLP-ORD-0283	Hub_3_Pune	25-07-2025 14:34
FLP-ORD-0134	Hub_3_Ahmedabad	26-08-2025 15:29
FLP-ORD-0241	Hub_3_Chennai	24-08-2025 12:31
FLP-ORD-0114	Hub_5_Hyderabad	29-08-2025 15:09
FLP-ORD-0214	Hub_3_Jaipur	21-08-2025 05:39
FLP-ORD-0126	Hub_5_Ahmedabad	26-07-2025 23:02
FLP-ORD-0251	Hub_2_Lucknow	31-07-2025 14:17
FLP-ORD-0280	Hub_4_Chennai	28-08-2025 17:41
FLP-ORD-0269	Hub_1_Hyderabad	30-08-2025 21:03

Insight

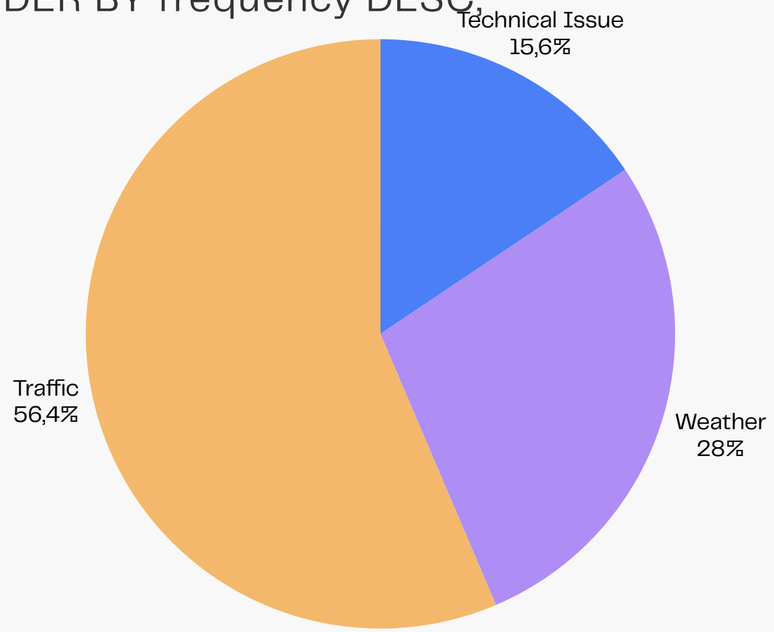
- Latest checkpoint shows current shipment status
- Helps track delays and identify where shipments are held up

TASK 6: SHIPMENT TRACKING ANALYTICS

COMMON DELAY REASONS

```
SELECT Delay_Reason,
       COUNT(*) AS frequency
FROM flipkart_shipmenttracking
WHERE Delay_Reason != 'None'
GROUP BY Delay_Reason
ORDER BY frequency DESC;
```

Delay_Reason	Frequency
Traffic	387
Weather	192
Technical Issue	107



Insight

- Certain delay reasons occur more frequently
- Indicates systematic issues (e.g., traffic, weather, operational delays)
- Helps prioritize problem areas

TASK 6: SHIPMENT TRACKING ANALYTICS

ORDERS WITH REPEATED DELAYED CHECKPOINTS

```
SELECT Order_ID,
      COUNT(*) AS delayed_checkpoints
FROM flipkart_shipmenttracking
WHERE Delay_Reason != 'None'
GROUP BY Order_ID
HAVING COUNT(*) > 2;
```

Order_ID	Delay_Frequency
FLP-ORD-0074	4
FLP-ORD-0190	5
FLP-ORD-0061	7
FLP-ORD-0134	4
FLP-ORD-0241	4
FLP-ORD-0114	7
FLP-ORD-0126	3
FLP-ORD-0251	7
FLP-ORD-0212	3
FLP-ORD-0221	5
FLP-ORD-0041	5
FLP-ORD-0152	3
FLP-ORD-0079	4
FLP-ORD-0286	4
FLP-ORD-0048	3
FLP-ORD-0125	4

Insight

- Some orders face multiple delays across checkpoints
- Indicates severe inefficiencies in shipment handling

KEY FINDINGS

01

Majority of shipments are on-time, but a subset experiences delays across checkpoints

02

Certain delay reasons occur more frequently, indicating recurring operational issues

03

Some orders face multiple delayed checkpoints, highlighting critical inefficiencies in the delivery process

TASK 7: ADVANCED KPI REPORTING

AVERAGE DELIVERY DELAY BY REGION

```
SELECT r.Start_Location,
                    AVG(DATEDIFF(o.Actual_Delivery_Date,    o.Order_Date))    AS
avg_delivery_time
FROM flipkart_orders o
JOIN flipkart_routes r
ON o.Route_ID = r.Route_ID
GROUP BY r.Start_Location;
```

Start_Location	avg_delay
Hyderabad	0.3806
Mumbai	0.4706
Pune	0.5556
Bengaluru	0.5000
Lucknow	0.5385
Ahmedabad	0.6061

Insight

- Delivery time varies across regions
- Some regions show consistently higher delays
- Indicates location-based inefficiencies

TASK 7: ADVANCED KPI REPORTING

ON-TIME DELIVERY PERFORMANCE

```
SELECT
   ROUND(
      (SUM(CASE
           WHEN Actual_Delivery_Date <= Expected_Delivery_Date THEN 1
           ELSE 0
         END) * 100.0 / COUNT(*)),
      2
   ) AS on_time_percentage
FROM flipkart_orders;
```

	On_Time_Percentage
▶	72.66667

Insight

- High on-time delivery % indicates strong performance
- Remaining delays highlight areas for improvement

TASK 7: ADVANCED KPI REPORTING

AVERAGE TRAFFIC DELAY PER ROUTE

```
SELECT Route_ID,
       AVG(Traffic_Delay_Min) AS avg_traffic_delay
FROM flipkart_routes
GROUP BY Route_ID;
```

Route_ID	Average_Traffic_Delay
RT_01	67.0000
RT_02	30.0000
RT_03	29.0000
RT_04	23.0000
RT_05	55.0000
RT_06	15.0000
RT_07	58.0000
RT_08	90.0000
RT_09	83.0000
RT_10	15.0000
RT_11	20.0000
RT_12	58.0000
RT_13	56.0000
RT_14	36.0000
RT_15	87.0000
RT_16	30.0000

Insight

- Some routes experience higher traffic delays
- Directly impacts delivery performance
- Requires traffic-aware planning

CONCLUSION

- The analysis identified key inefficiencies across routes, warehouses, and delivery agents
- Traffic delays and poor route efficiency are major contributors to delivery delays
- Certain warehouses act as bottlenecks, impacting overall processing and delivery time
- Agent performance varies, but speed alone is not the primary factor affecting delivery outcomes
- Optimizing routes, workload distribution, and operational processes can significantly improve delivery efficiency

Data-driven optimization of logistics operations can enhance delivery performance, reduce delays, and improve overall customer satisfaction.

Presented by :
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THANK YOU

FOR YOUR TIME
AND ATTENTION

Presented by :
SUJAL AGRAHARI

video Links: Presentation Part 1
Presentation Part 2